

GRE QUANT MASTER SHEET — 170 STRATEGY BATTLE MAP

SECTION 1 — NUMBERS & PROPERTIES

TYPES OF NUMBERS

- **Natural:** >0 (1, 2, 3...) | **Whole:** ≥ 0 (0, 1, 2...)
- **Integers:** ..., -2, -1, 0, 1, 2...
- **Positive:** >0 | **Negative:** <0
- **Non-Positive:** ≤ 0 | **Non-Negative:** ≥ 0
- **Rational:** Fractions p/q | **Irrational:** non-terminating (e.g., $\sqrt{2}$, π)
- **Real:** All points on the number line.
- **Even:** Divisible by 2 ($2k$) | **Odd:** Not divisible ($2k+1$)
- **Prime:** Only 2 factors (1 & itself). (2, 3, 5, 7...) *2 is only even prime.
- **Composite:** >2 factors. (4, 6, 8, 9...) *1 is neither.

PLACE VALUE

Number 5,432: 5(Thousands), 4(Hundreds), 3(Tens), 2(Units).
Digit Value: True numeric value. *Face Value:* The digit itself.

DIVISION ALGORITHM

Dividend = *Divisor* $\times$ *Quotient* + *Remainder*

Constraint: $0 \leq \text{Remainder} < \text{Divisor}$

LEAST NUMBER QUESTIONS (DIVISIBILITY BY Y)

- Least to add to X: $Y - (X \bmod Y)$
- Least to subtract from X: $X \bmod Y$

DIVISIBILITY RULES

#	Rule	#	Rule
2	Last digit is even	6	Divisible by 2 AND 3
3	Sum of digits div by 3	8	Last 3 digits div by 8
4	Last 2 digits div by 4	9	Sum of digits div by 9
5	Last digit 0 or 5	10	Last digit is 0
11: Diff between sum of odd-placed and even-placed digits is 0 or div by 11.			

FACTORS, MULTIPLES & PRIMES

If $N = p^a \cdot q^b \cdot r^c$ (prime factorization):

- **Number of factors:** $(a+1)(b+1)(c+1)$
- **LCM $\times$ GCF:** *Product of two numbers* ($A \times B$)

EVEN/ODD PROPERTIES

$E \pm E = E$ $O \pm O = E$ $E \pm O = O$	$E \times E = E$ $O \times O = O$ $E \times O = E$	$E^n = E$ ($n > 0$) $O^n = O$ ($n > 0$)
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SECTION 2 — PROGRESSIONS & SEQUENCES

ARITHMETIC PROGRESSION (AP)

n th term: $T_n = a + (n-1)d$

Sum of n terms: $S_n = \frac{n}{2} [2a + (n-1)d]$

If a, b, c are in AP: $2b = a + c$ (Arithmetic Mean)

GEOMETRIC PROGRESSION (GP)

n th term: $T_n = a \cdot r^{n-1}$

Sum: $S_n = a(r^n - 1) / (r - 1)$

If a, b, c are in GP: $b^2 = a \cdot c$ (Geometric Mean)

SPECIAL SUMS

- $1 + 2 + \dots + n = n(n+1)/2$
- $1^2 + 2^2 + \dots + n^2 = n(n+1)(2n+1)/6$
- $1^3 + 2^3 + \dots + n^3 = [n(n+1)/2]^2$

SECTION 3 — PERCENTS

x% of y = $(x/100) \cdot y$ = y% of x

- **Percent Increase:** $(\text{Increase} / \text{Original}) \times 100$
- **Percent Decrease:** $(\text{Decrease} / \text{Original}) \times 100$
- **What % More/Less?** $|\text{Difference}| / \text{Base Value} \times 100$

MULTIPLIER METHOD

x% Increase $\rightarrow$ Mult by $1 + x/100$

x% Decrease $\rightarrow$ Mult by $1 - x/100$

SUCCESSIVE PERCENTAGE CHANGE

Total % Change = $x + y + (xy/100)$

IF Equal Increase (x%) then Decrease (x%) $\rightarrow$ THINK Net Loss = $x^2/100$ %

SECTION 4 — EXPONENTS & ROOTS

LAWS OF EXPONENTS

$$a^m \times a^n = a^{m+n}$$

$$a^m / a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$a^0 = 1 \text{ (} a \neq 0 \text{)}$$

$$a^{-n} = 1 / a^n$$

LAWS OF ROOTS

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

$$\sqrt{a/b} = \sqrt{a} / \sqrt{b}$$

Important Trap:

$$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$$

IF Even root (e.g., $\sqrt{x^2}$) $\rightarrow$ THINK $|x|$ (Always Non-Negative)

SECTION 5 — ALGEBRA & INEQUALITIES

IDENTITIES & FORMULAS

- Diff of Squares: $a^2 - b^2 = (a-b)(a+b)$
- Perfect Squares: $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- Quad: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

FUNCTIONS

$f(x), g(x)$. Composite: $f(g(x))$ (Work inside-out).

INEQUALITIES & ABSOLUTE VALUE

- **Rule:** Multiplication/Division by a negative number **reverses** the inequality sign.
- **Compound:** $a < x < b$. Solve all parts simultaneously.
- $|x| = a \rightarrow x = a$ OR $x = -a$
- $|x| < a \rightarrow -a < x < a$ (Distance from 0 $< a$)
- $|x| > a \rightarrow x > a$ OR $x < -a$

IF Variable in Inequality $\rightarrow$ THINK Test: Positive, Negative, Zero, Fraction

SECTION 6 — RATE, WORK, SPEED & INTEREST

DISTANCE, SPEED, TIME

$$D = S \times T \mid S = D / T \mid T = D / S$$

Relative Speed:

Opposite Dir: Add Speeds ($S_1 + S_2$)

Same Dir: Subtract Speeds ($|S_1 - S_2|$)

WORK

$$\text{Work} = \text{Rate} \times \text{Time}$$

$$\text{Combined Work: } 1/T_{\text{total}} = 1/A + 1/B$$

INTEREST

Simple (SI): $PRT / 100$

Compound (CI): $\text{Amount} = P(1 + R/100n)^{nt}$
(n = periods per year, e.g. Half-Yearly: n=2, R/2, 2t)

SECTION 7 — GEOMETRY

LINES & ANGLES

Straight Angle = 180° | Around Point = 360°

Vertically Opposite = Equal | Linear Pair = 180°

POLYGONS (N SIDES)

Metric	Any Polygon	Regular Polygon
Sum Int. Angles	$(n-2) \times 180^\circ$	N/A
Sum Ext. Angles	360°	N/A
Each Int. Angle	N/A	$[(n-2)180^\circ]/n$
Each Ext. Angle	N/A	$360^\circ/n$

SECTION 8 — TRIANGLES

CORE PROPERTIES

- Angle Sum = 180°
- Ext Angle = Sum of 2 opposite interior angles.
- **Inequality:** Sum of any 2 sides $>$ 3rd side.
- **Inequality:** Diff of any 2 sides $<$ 3rd side.
- Equal Sides $\leftrightarrow$ Equal Opposite Angles.

SIMILARITY & CONGRUENCE

Similarity: AA, SAS, SSS (Sides proportional).

Congruence: SSS, SAS, ASA, AAS, RHS (Sides equal).

AREA FORMULAS

Standard: $\frac{1}{2} b h$

Heron's: $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$

Equilateral: $(\sqrt{3}/4)a^2$

RIGHT TRIANGLES (PYTHAGORAS $A^2+B^2=C^2$)

Triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25

Special Triangle	Ratio (Sides)
$30^\circ - 60^\circ - 90^\circ$	$1 : \sqrt{3} : 2$
$45^\circ - 45^\circ - 90^\circ$	$1 : 1 : \sqrt{2}$

SECTION 9 — QUADRILATERALS

Angle Sum = 360°

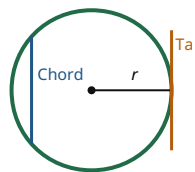
Shape	Properties / Area	Diagonals
Square	a^2 . All sides/angles equal.	Equal, bisect @ 90°
Rectangle	L $\times$ W. Opposite sides equal.	Equal, bisect
Rhombus	$\frac{1}{2}(d_1 \times d_2)$. All sides equal.	Bisect @ 90°
Parallelogram	b $\times$ h. Opp sides parallel.	Bisect each other
Trapezium	$\frac{1}{2}(a+b)h$. 1 pair parallel.	-

Every square is a rectangle, but not every rectangle is a square. Every rhombus is a parallelogram.

SECTION 10 — CIRCLES

Circumference: $2\pi r$ | Area: πr^2

- **Arc Length:** $(\theta/360) \times 2\pi r$
- **Sector Area:** $(\theta/360) \times \pi r^2$
- Area from Arc: $\frac{1}{2} \times r \times \text{Arc Length}$



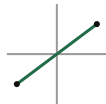
Angles: Angle subtended at center (2θ) is double the angle subtended at circumference (θ).

SECTION 11 — COORDINATE GEOMETRY

Slope (m): $(y_2 - y_1) / (x_2 - x_1)$

Distance: $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Midpoint: $[(x_1 + x_2)/2, (y_1 + y_2)/2]$



Parallel lines: $m_1 = m_2$

Perpendicular lines: $m_1 \times m_2 = -1$

SECTION 12 — 3D MENSURATION

Shape	Volume	Surface Area (Total)
Cube	a^3	$6a^2$ (Diag: $a\sqrt{3}$)
Cuboid	$l \cdot w \cdot h$	$2(lw + wh + hl)$
Cylinder	$\pi r^2 h$	$2\pi r(r + h)$
Cone	$\frac{1}{3} \pi r^2 h$	$\pi r(r + l)$ [l =slant]
Sphere	$\frac{4}{3} \pi r^3$	$4\pi r^2$
Hemisph.	$\frac{2}{3} \pi r^3$	$3\pi r^2$

SECTION 13 — AVERAGES & WORD PROBLEMS

Average = *Sum* / *Count*

Weighted Average: $(w_1x_1 + w_2x_2) / (w_1 + w_2)$

Mixtures: Use weighted average logic for concentration blends.

TRANSLATING STATEMENTS

"20% as many men as women" → $M = 0.20W$

GRE Tip: If answer choices are integers → Assume Variables. If choices are fractions/percents → Assume Numbers (e.g., 100).

SECTION 14 — STATISTICS

- **Mean:** Arithmetic average.
- **Median:** Middle value (sort first!). If even count, average of middle two.
- **Mode:** Most frequent value.
- **Range:** Max – Min.
- **Quartiles / Quintiles:** Divides data into 4 / 5 equal parts.
- **IQR:** Q3 – Q1.



SECTION 15 — PROBABILITY

P(E) = Favorable Outcomes / Total Outcomes

Complement: $P(\text{Not } E) = 1 - P(E)$

Addition Rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

Mutually Exclusive: $P(A \cap B) = 0$

Independent Events: $P(A \text{ and } B) = P(A) \times P(B)$

Neither Event: $1 - P(A \text{ or } B)$

SECTION 16 — COUNTING

Fund. Counting Principle: Multiply options for independent stages.

Permutation (Order matters):

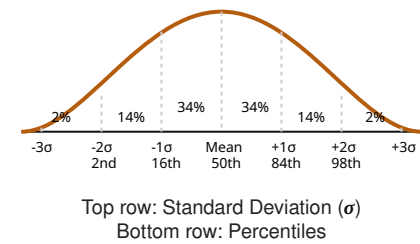
${}^nP_r = n! / (n-r)!$

Combination (Order doesn't matter):

${}^nC_r = n! / [r!(n-r)!]$

Choice Logic: Total subset combinations of n items = 2^n

SECTION 17 — NORMAL DISTRIBUTION



SECTION 18 — GRE MASTER TRAPS

DO NOT ASSUME FIGURES ARE DRAWN TO SCALE.

Exceptions (usually drawn to scale):

- Data Interpretation Graphs
- Coordinate Geometry

Always Test:

Positive	Negative	Zero	Fraction
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Check: Units (mins vs hours), Restrictions (x is an integer), Hidden Geometry Patterns.

Use Problem-Solving Frameworks:

- Plugging in Numbers
- Backsolving (Using answer choices)
- Estimation